

Information Systems Management

Chapter # 11

Managing Knowledge and Artificial Intelligence

Case Study Presentation

#1 and #2

What is the Role of Knowledge Management Systems in Business?

Knowledge management systems among fastest growing areas of software investment

Information economy: production and distribution of information and knowledge a major source of wealth and prosperity

Substantial part of a firm's stock market value is related to intangible assets: knowledge, brands, reputations, and unique business processes

Well-executed knowledge-based projects can produce extraordinary R O I

Important Dimensions of Knowledge

- Data, information, knowledge, and wisdom
- Tacit knowledge and explicit knowledge
- Important dimensions of knowledge
 - Knowledge is a firm asset
 - Knowledge has different forms
 - Knowledge has a location
 - Knowledge is situational

Important Dimensions of Knowledge

- Knowledge-based core competencies
 - Key organizational assets
- Knowing how to do things effectively and efficiently in ways others cannot duplicate is a prime source of profit and competitive advantage
 - Example: Having a unique build-to-order production system
- Organizational learning
 - Process in which organizations gain experience through collection of data, measurement, trial and error, and feedback

Knowledge Management Value Chain (1/3)

Knowledge management

- Set of business processes developed in an organization to create, store, transfer, and apply knowledge

Knowledge management value chain

- Each stage adds value to raw data and information as they are transformed into usable knowledge
 - Knowledge acquisition
 - Knowledge storage
 - Knowledge dissemination
 - Knowledge application

Knowledge Management Value Chain (2/3)

Knowledge acquisition

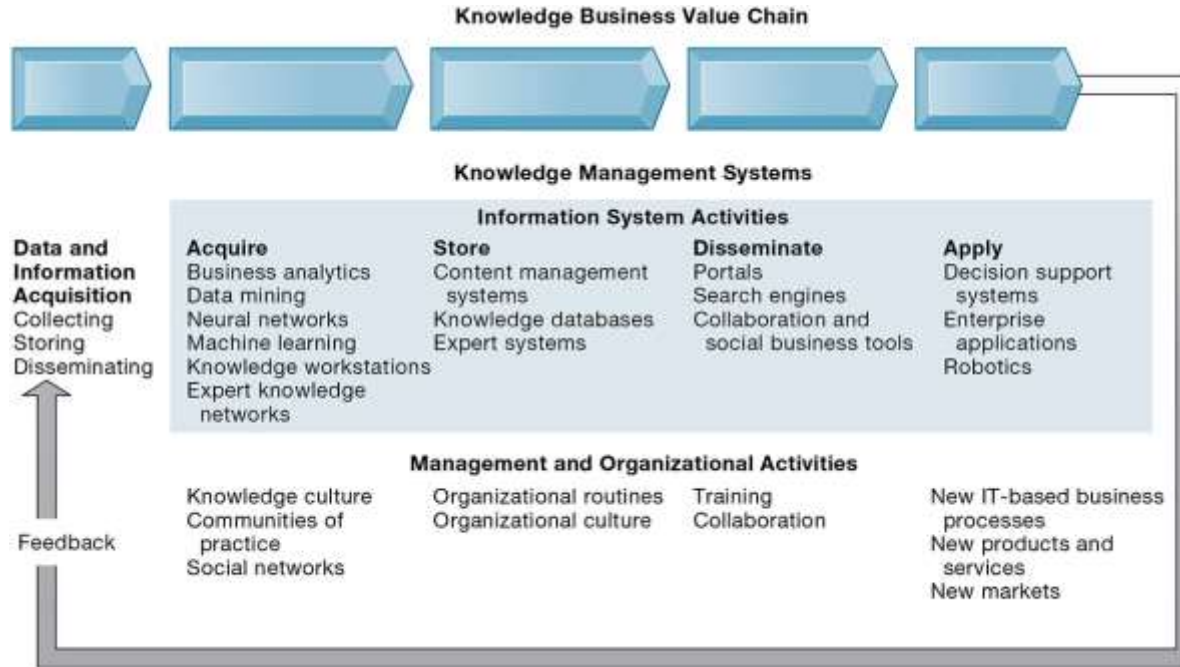
- Documenting tacit and explicit knowledge
 - Storing documents, reports, presentations, best practices
 - Unstructured documents (e.g., e-mails)
 - Developing online expert networks
- Creating knowledge
- Tracking data from T P S and external sources

Knowledge storage

- Databases
- Document management systems
- Role of management

Knowledge Management Value Chain (3/3)

- Knowledge dissemination
 - Portals, wikis
 - E-mail, instant messaging
 - Search engines, collaboration tools
 - A deluge of information
 - Training programs, informal networks, and shared management experience help managers focus attention on important information.
- Knowledge application
 - New business practices
 - New products and services
 - New markets

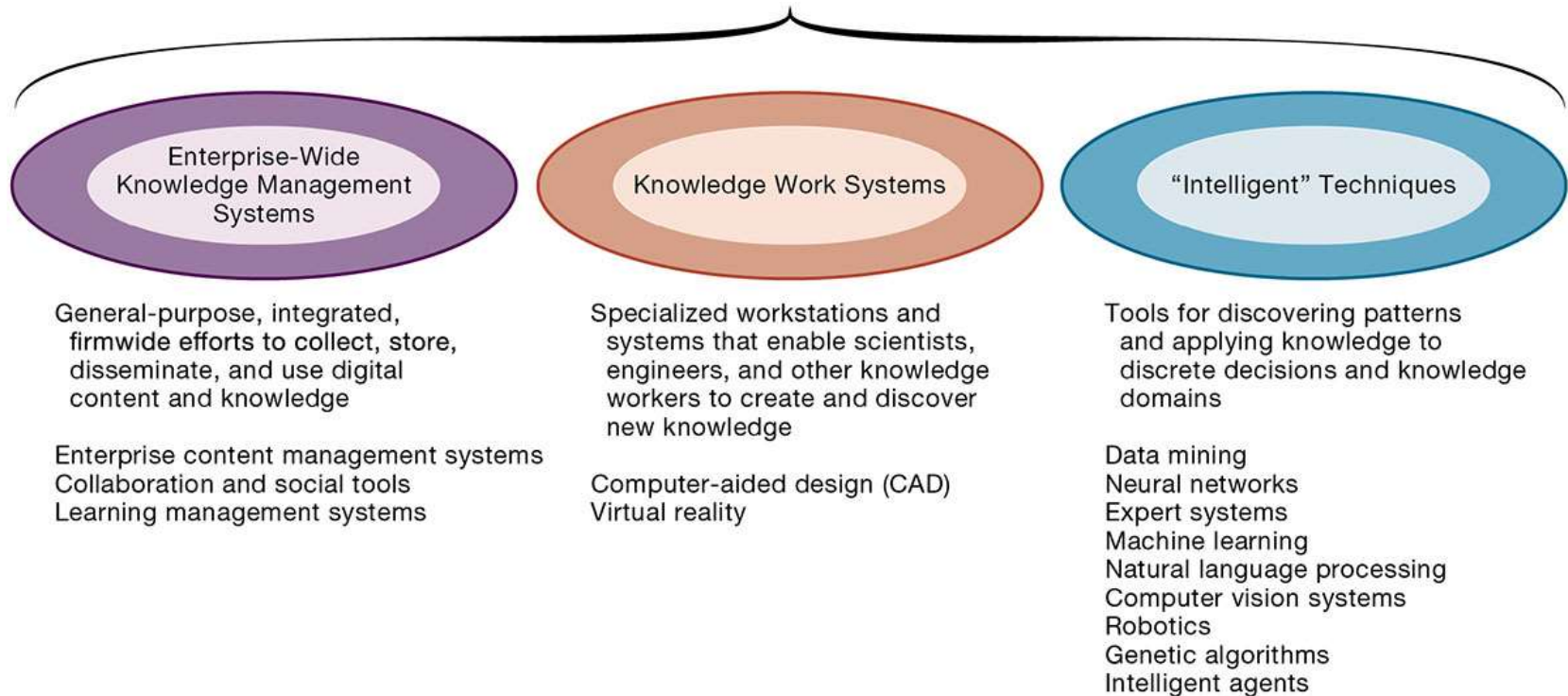


The Knowledge Management Value Chain

Types of Knowledge Management Systems

- Enterprise-wide knowledge management systems
 - General-purpose firm-wide efforts to collect, store, distribute, and apply digital content and knowledge
- Knowledge work systems (KWS)
 - Specialized systems built for engineers, scientists, other knowledge workers charged with discovering and creating new knowledge
- Intelligent techniques
 - Diverse group of techniques, such as data mining, expert systems, machine learning, used for various goals: discovering knowledge, distilling knowledge, discovering optimal solutions

Major Types of Knowledge Management Systems



What are AI and ML

Intelligent techniques are often described as Artificial Intelligence

Foundations of “Human Intelligence” or “Common Sense”

- Humans see, hear, communicate, plan, achieve goals and perceive patterns and learn
- Humans also love, hate and choose what they want to pursue

A narrow definition of AI is more realistic and possible.

- computer programs **BUT** capable of performing far complex tasks
- Interpreting CT scans, recognizing faces and voices, playing games like chess or Go, or besting human experts in certain well-defined tasks
- In industries they are transforming how business is done, where people are employed, and how they do their jobs.

Evolution of AI

Forces :

- Development of Big Data databases (Internet, e-commerce, IoT, and social media)
- Drastic reduction in the cost of processing and the growth in the power of processors.
- Refinement of algorithms by AI engineers and researchers
- Significant investment from business and governments.

In last 75 yrs, few fundamental conceptual breakthroughs in AI happened or in understanding how humans think.

- Algorithms and statistical techniques were developed decades earlier but could not be implemented and refined on such a large scale as is currently possible.

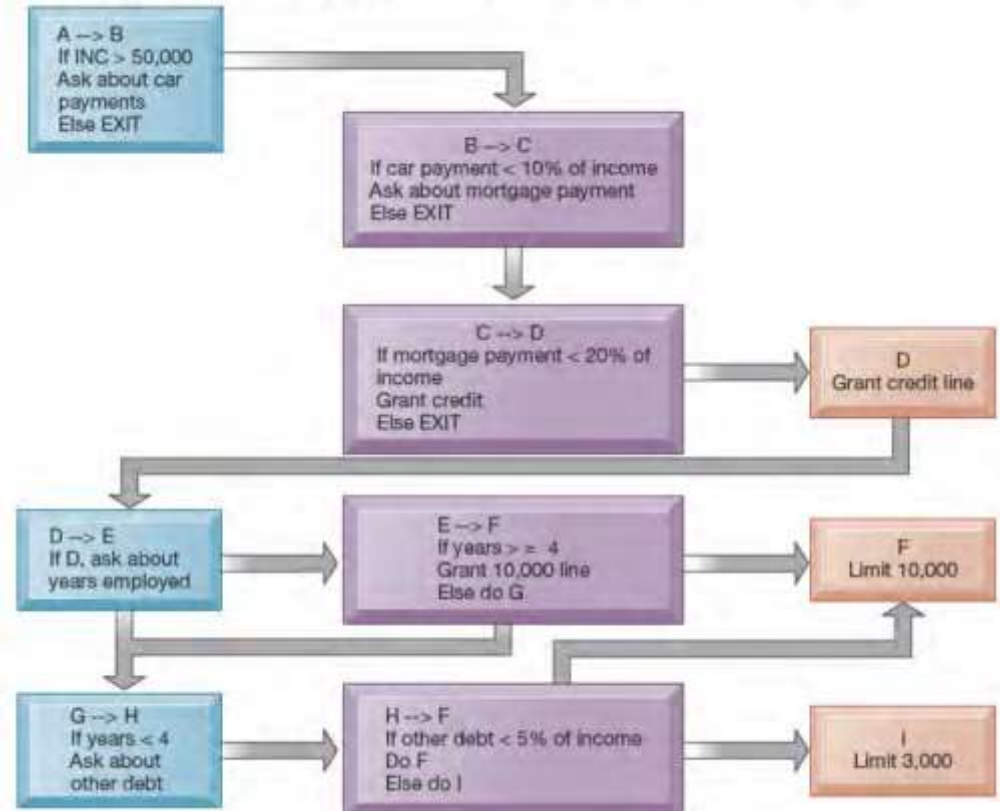
In a famous **1950 paper**, computer scientist **Alan Turing** defined an artificially intelligent computer program as one that a human could have a conversation with and not be able to tell it was a computer

Major types of AI Techniques

Expert systems	Represent the knowledge of experts as a set of rules that can be programmed so that a computer can assist human decision makers.
Machine learning	Software that can identify patterns in very large databases without explicit programming although with significant human training.
Neural networks and deep learning	Loosely based on human neurons, algorithms that can be trained to classify objects into known categories based on data inputs. Deep learning uses multiple layers of neural networks to reveal the underlying patterns in data, and in some limited cases identify patterns without human training.
Genetic algorithms	Algorithms based loosely on evolutionary natural selection and mutation, commonly used to generate high-quality solutions to optimization and search problems.
Natural language processing	Algorithms that make it possible for a computer to understand and analyze natural human language.
Computer vision systems	Systems that can view and extract information from real-world images.
Robotics	Use of machines that can substitute for human movements as well as computer systems for their control and information processing.
Intelligent agents	Software agents that use built-in or learned knowledge to perform specific tasks or services for an individual.

Expert Systems

- Capture the knowledge of experts and represent that knowledge as set of rules.
- Set of rules converted into IF-THEN rules.
- ~ **20%** of the AI systems today are Expert Systems



Machine Learning

- Used by neural networks, deep learning networks, and genetic algorithms
- Different paradigm than expert systems
- Focuses on recognizing patterns in very large data sets
- Contemporary examples
 - Facebook ad display
 - Netflix recommender system

Machine Learning

- Supervised learning
 - System “trained” on input-output pairs (labeled data). Goal is to map input to a known output identified by humans.
 - Tasks : Classification & Regression
 - Eg :
 - SPAM detection
 - Image classification,
 - Predicting weather patterns, likelihood of disease
 - Sentiment analysis

Machine Learning

- Unsupervised learning
 - Algorithms are trained on unlabeled data, finding hidden patterns or structures without prior guidance
 - Tasks : Clustering, Associations, Anomaly detection
 - Eg :
 - Grouping customers with similar purchasing habits, behaviors, or demographics
 - Analyzing user behavior and product relationships to suggest relevant items
 - Identifying unusual patterns or outliers in data that could indicate fraudulent transactions,

Machine Learning

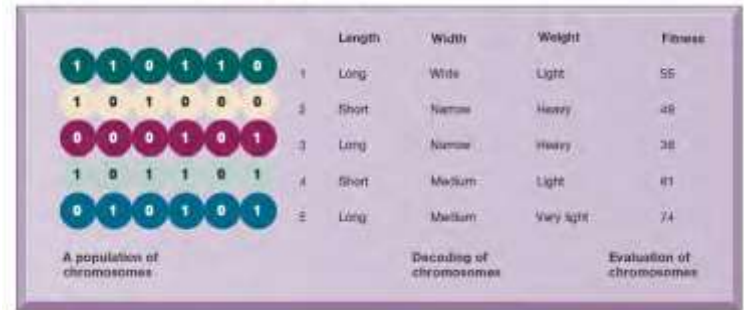
- Reinforcement Learning
 - Systems learn through trial-and-error by interacting with an environment to maximize cumulative rewards.
 - Tasks : Sequential decision-making, game playing, robotics
 - Eg :
 - Teaching a robot to walk,
 - training AI for Chess, Go
 - Developing autonomous cars
 - Aligning language models like ChatGPT with human preference and safer conversation

Neural Networks

- Neural network is composed of interconnected units called neurons. Each neuron can take data from other neurons, and transfer data to other neurons.
- Are **pattern detection programs**. Learn patterns from large quantities of data and ultimately finding pathways
- Finding applications in medicines and business
 - Diagnosing epilepsy/ Alzheimer's
 - Credit-card fraud for sudden changes in buying patterns

Genetic Learning

- Another form of M/L
- Examine a large number of alternative solutions and find an optimal solution
- Inspired by evolutionary biology (inheritance, mutation, selection, crossover etc)
- Eg optimize the design for jet turbine aircraft engines,



NLP, Computer Vision, Robotics

Natural language processing (NLP) makes it possible for a computer to understand and analyze natural language—language that human beings instinctively use

Computer vision systems deal with how computers can emulate the human visual system to view and extract information from real-world images.

Robotics deals with the design, construction, operation, and use of movable machines that can substitute for humans along with computer systems for their control, sensory feedback, and information processing

Intelligent Agents

Intelligent agents are software programs that work in the background without direct human intervention to carry out specific tasks for an individual user, business process, or software application.

The agent uses a limited built-in or learned knowledge base to accomplish tasks or make decisions on the user's behalf,

Case Study Presentation

#3 and #4

What Types of Systems Are Used for Enterprise-Wide Knowledge Management?

Three major types of knowledge in an enterprise

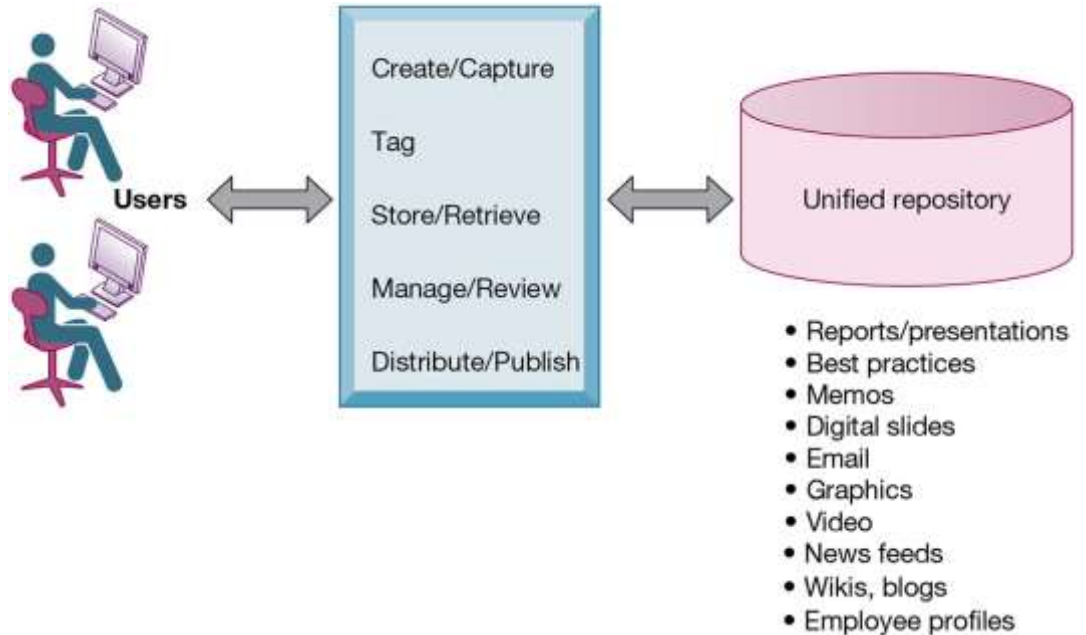
- Structured documents
 - Reports, presentations
 - Formal rules
- Semistructured documents
 - E-mails, videos
- Unstructured, tacit knowledge

80% of an organization's business content is semistructured or unstructured

Enterprise Content Management Systems

- Help capture, store, retrieve, distribute, preserve documents and semistructured knowledge
- Bring in external sources
 - News feeds, research
- Tools for communication and collaboration
 - Blogs, wikis, and so on
- Key problem: developing taxonomy
- Digital asset management systems

An Enterprise Content Management System



Learning Management Systems (LMS)

Provide tools for management, delivery, tracking, and assessment of employee learning and training

Support multiple modes of learning

- web-based classes, online forums, and so on

Automates selection and administration of courses

Assembles and delivers learning content

Measures learning effectiveness

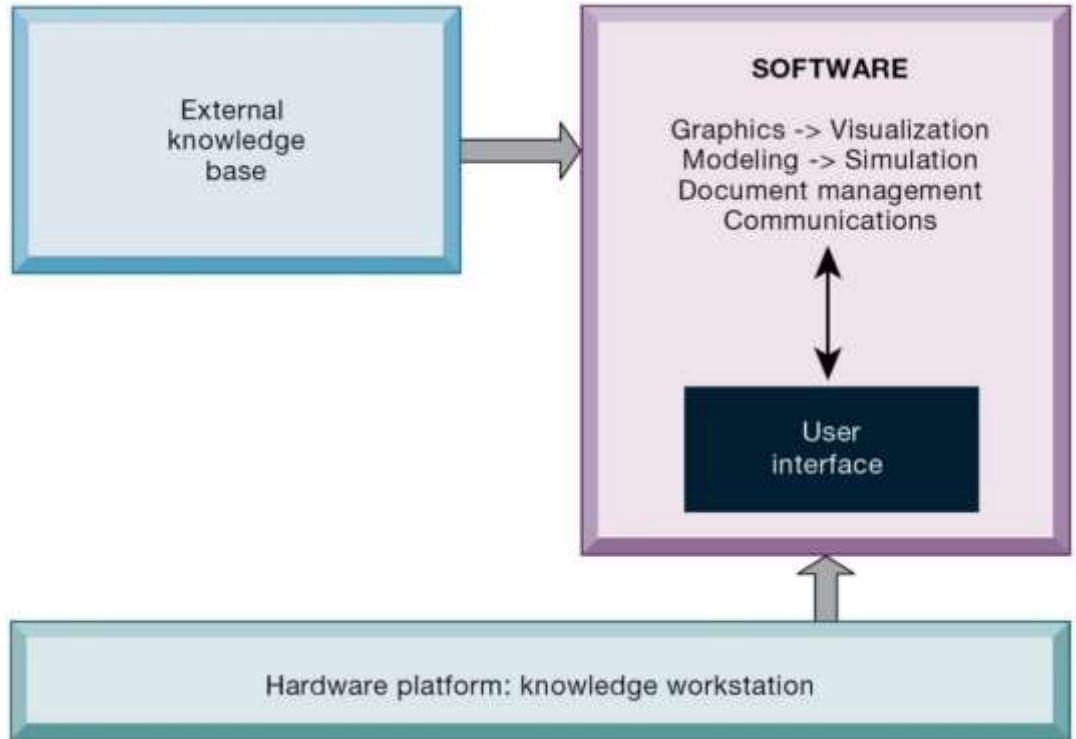
Massively open online courses (M O O C s)

- Web course open to large numbers of participants

Major types of Knowledge work systems and how they provide value for firms.

- **Knowledge Workers and Knowledge Work**
- Knowledge workers
 - Researchers, designers, architects, scientists, engineers who create knowledge for the organization
 - Perform key roles critical to organization and managers who work within organization
- Knowledge work systems
 - Systems for knowledge workers to help create new knowledge and integrate that knowledge into business

Requirements of KMS



Examples of Knowledge Work Systems

- CAD (computer-aided design)
 - Creation of engineering or architectural designs
 - 3D printing
- Virtual reality systems
 - Simulate real-life environments
- Augmented reality (AR) systems
 - Enhance visualization by overlaying digital data and images onto physical real-world environment

Feature	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Data Type	Uses labeled data (input-output pairs).	Uses unlabeled data .	Learns through interactions with an environment.
Primary Goal	To map inputs to a known output identified by humans.	To find hidden patterns or structures without prior guidance.	To maximize cumulative rewards through trial-and-error.
Common Tasks	Classification and Regression.	Clustering, Associations, and Anomaly detection.	Sequential decision-making and game playing.
Typical Example	SPAM detection or sentiment analysis.	Grouping customers with similar purchasing habits.	Training AI for Chess, Go , or autonomous cars.